Hertie School

Where Bike Thou? A Machine Learning Approach to Demand Prediction and Bikesharing Station Prioritization in Mexico City

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"At present, cyclists don't have much clout in pushing for the facilities and city qualities they need and want. I think they would have more clout if they pedaled along with their many, many potential allies, getting aid from those allies in support of cyclists' needs, and in return helping their natural allies in their battles for better quality of city life: working in mutual support with people who care about bullfrogs, or about traffic lights where the schoolchildren cross, or about threats of expropriation to their working places or their homes, understanding that the specific needs and desires of cyclists can be furthered only within a broader context of the city as a decent place for people to live and get around in."
—Jane Jacobs (2017) in "Pedaling Together" from *Vital Little Plans*

Abstract

Can learning from historical bike flows help predict future availability needs? Bicycle-sharing systems (BSS) have been proposed as solutions to urbanization, congestion, and mobility inequalities. These challenges are particularly acute in urban areas of Latin America, a region that faces important fiscal, infrastructure, and development constraints. This thesis addresses whether demand prediction can inform station prioritization for potential rebalancing strategies, using seven years of operational data from ECOBICI, Mexico City's BSS and the largest in the region. A machine learning framework with time-based splits is implemented to predict station-level hourly departures and identify stations driving system imbalance. Among the models evaluated, including Linear, Poisson, and tree-based models, XGBoost achieves the lowest error metrics (MAE: 2.285, RMSE: 3.441) but underestimates peak demand periods where prioritization matters most.

Predicted demand and absolute imbalance are then integrated into a prioritization framework, applied separately across weekday and weekend periods. Results show that prediction-informed prioritization captures 12.3% of total imbalance, compared to 14.6% based on observed 2025 imbalance. However, the framework identifies a different station subset driving this imbalance, with only 41% overlap compared to the baseline. These models do not outperform historical rankings, yet they offer an alternative lens for rebalancing strategies under operational constraints such as station capacity. This thesis contributes to the underrepresented literature on Latin American BSS by leveraging operational data to inform planning and accountability in resource allocation, highlighting the importance of understanding where and why predictions diverge from system behavior.

Key words: bike-sharing, prediction, machine learning, prioritization, Mexico City

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Chapter 1

Introduction

1.1 Background and Motivation

Urban mobility systems have transformed over the past forty years as cities face the pressures of urbanization, congestion, environmental degradation, and persisting social inequalities. These challenges are particularly acute in large metropolitan areas of Latin America, such as Mexico City, where fiscal, infrastructure and development constraints limit the access to efficient and affordable mobility options for all, especially peripheral and disadvantaged areas (Larrañaga et al. 2025). In this context, cycling has become a sought-after alternative to reduce car dependency, alleviate parking space, and promote both personal and shared mobility (Shaheen et al. 2010; Calatayud et al. 2021).

Bicycle-sharing systems (BSS) are such a solution and consist of a network of publicly available bicycles placed across docking stations or free-floating areas, ready to be used for low payment (Fishman 2016; Vallez et al. 2021). Like other shared mobility services, BSS can be understood as quasi-public goods reshaping urban mobility and generating collective benefits, while also bringing new political, regulatory and transport implications (Bartling 2019). Empirical evidence suggests that BSS can improve first- and last-mile connectivity, low-carbon mobility, and increase access, particularly when integrated with public transport (Griffin et al. 2016; Wang and Zhou 2017; Ashqar et al. 2020). Currently, BSS operate in over 3,000 cities worldwide (Thompson et al. 2025), yet their benefits depend on ensuring that bikes are available when and where users need them.

1.2 The Balancing Problem in BSS

This thesis focuses on a core issue of BSS: the spatial and temporal imbalance in bike availability. As users move across the city, some stations will accumulate bikes (surplus) while others will experience shortages (deficit) during service hours. This is known in the literature as the balancing or repositioning problem (Vallez et al. 2021; Maya-Duque et al. 2022). In cities marked by urban and spatial inequalities, these imbalances reinforce disparities in access to mobility, making the problem not only operational but also socially relevant to inform the distribution of bikes across the BSS.

1.3 Mexico City as a BSS Case Study

Mexico City is a relevant and compelling BSS case to examine the balancing problem. As one of the largest metropolitan areas in the world, it concentrates 22 million people and 15% of the national economy (CONAPO 2024; INEGI 2025). Due to its scale, density, and overlapping urban challenges, the management of its transport systems serves as a point of reference for other cities and regions (Rivera et al. 2011).

The city is also home to the largest BSS in Latin America: ECOBICI ("Sistema de Transporte Individual en Bicicleta Pública ECOBICI"). Launched in 2010 as part of a broader strategy to reduce congestion and emissions, ECOBICI is a docked BSS in which registered users take and return bikes at fixed stations, accessing the network through a card-based annual subscription priced at US$16; its objective is to promote active mobility as a viable and safe alternative within a transport model historically dominated by private car use (UNDESA n.d.; Romero Sánchez 2010; Delgado Peralta 2017).

Since its launch, ECOBICI has evolved from a small network into a system with over 9,300 bikes and nearly 700 stations, with an estimated 180,000 monthly active users (Mexico City Secretary of Mobility 2025). This growth has taken place over different policy phases, including a major expansion and modernization process after the COVID-19 pandemic, and this has increased coverage to more neighborhoods and reduced costs (Open Contracting Partnership 2024). While expansion has improved access, demand is not evenly distributed, concentrating around specific stations and during peak periods.

These dynamics call for a more rigorous analysis given the highly unequal context in which ECOBICI operates. Disparities in infrastructure, maintenance and travel conditions continue to shape mobility choices (Rivas et al. 2025); at the same time, ECOBICI itself faces operational constraints: some bikes are used at a very high frequency, 14% of fleet may be out of service due to intensive usage, and human resources for maintenance and rebalancing remain limited (Arana 2025; García Nieto 2025; Gutiérrez 2025; Navarrete 2026). Addressing this requires an analytical framework to anticipate demand and prioritize stations with the most need in an already constrained network.

1.4 Related Works and Research Gap

Previous BSS research has, for the most part, largely focused on the operations behind rebalancing. These are divided between demand prediction and optimization rebalancing approaches, with important contributions and progress yet limited in their joint assessment and application within Latin American contexts.

The first strand of literature framed rebalancing as a routing and inventory optimization task. Chemla et al. (2013) provide one of the first definitions of rebalancing, modeling it as a pick-up-and-delivery problem under capacity constraints, where the objective is to minimize both surpluses and deficits by moving bikes from one station to another. Further research extends this definition by adding service levels and user dissatisfaction. For instance, mixed-integer linear programming has been applied to propose an optimal bike allocation and capture imbalance through user penalties (Raviv et al. 2013), whereas stochastic modeling with Markov chains has proposed that inventory levels should use ranges rather than fixed targets (Schuijbroek et al. 2017).

Additional contributions by Gómez Márquez et al. (2021) improve these models by jointly optimizing routing and loading choices integrating service time and minimum demand satisfaction. These methods are mathematically rigorous and aligned with operations decision-making, yet they tend to rely on simplified or static assumptions of demand or become computational infeasible as they increase in size, limiting their scalability in large urban settings. Attempts to address the first challenge included simulations or scenario-based optimization that integrate uncertainty, temporal dynamics and penalties for unmet demand (Kloimüllner et al. 2014; Dell'Amico et al. 2018). These approaches reflect real-world conditions and capture variability, but depend on strong assumptions about demand distributions and involve high computational costs.

The second strand of literature focuses on machine learning or advanced computational methods. These approaches use heterogeneous data sources, ranging from temporal, spatial and environmental features, as part of a strategy to improve the accuracy of predictions and bike volumes (Ashqar et al. 2020; Miah et al. 2023; Kaiser et al. 2024). While these models enhance the understanding of demand patterns, they do not consider rebalancing or are developed independently of a potential optimization in a docked system. In particular, Ashqar et al. (2020)'s work is relevant for this thesis as it applies Random Forest and boosting techniques to improve the prediction of bike availability.

More recent work applied to Mexico City's ECOBICI system addresses the uneven contribution of stations to imbalance by implementing clustering, districting and priority rankings. Maya-Duque et al. (2022) propose a model that groups stations into districts based on demand patterns and operational constraints, stressing that not all stations contribute equally to measures of imbalance. This approach, too, inspires part of the prioritization component of this thesis as it introduces a measure of station-level prioritization, or ranking, as seen in Chapter 3. Similarly, simulation-based studies using non-homogenous Poisson processes show that shortages in a BSS can be reduced by combining demand modeling and inventory optimization (Gómez Márquez et al. 2021).

Despite progress in the field, the literature remains fragmented and most applications are largely concentrated in North America, Europe and Asia, with fewer cases using advanced models on Latin American cities (Vallez et al. 2021; Larrañaga et al. 2025). Lastly, most of the BSS literature has tackled either demand prediction or optimization, and not on a unified framework considering measures of imbalance and heterogeneous features.

1.5 Research Questions and Objectives

This thesis addresses the rebalancing problem in bike-sharing systems by integrating demand prediction and imbalance magnitude, using ECOBICI as a case study. Concretely, the thesis seeks to answer:

- ***In what ways and how well can demand prediction inform station prioritization for identifying high-impact rebalancing targets in bicycle-sharing systems?***

To address the question, the thesis first seeks to test and compare methods to predict station-level demand, using historical records and contextual features. It evaluates the performance of linear models and tree-based models to predict departures in a station-level panel. Second, it identifies high-demand stations and peak periods that disproportionately drive system imbalance. Third, it integrates prediction results with a prioritization framework targeting critical stations, assessing imbalance patterns and system performance under operational constraints.

1.6 Scope and Contributions

The scope of this thesis is limited to demand prediction and station prioritization in a docked BSS system. The analysis is conducted in a historic, offline setting and does not provide a real-time model or routing recommendations. A primary contribution of this thesis is the application of these methods to Mexico City's ECOBICI, a large network with documented imbalance challenges. A secondary contribution is to assess how demand prediction could best provide new information to inform prioritization for rebalancing and resource allocation that advance transport access in similar urban contexts in Latin America (Larrañaga et al. 2025).

Chapter 2

Data

2.1 Data Source Description

Mexico City's Secretary of Mobility publishes ECOBICI (2022) system data through its dedicated open data portal, providing user demographics, bike and station identifiers, and trip-level information (start date, start time, end station, end date and end time) in JSON format. Historical trip records are available as monthly CSV files with one month's delay, while real-time system data is accessible through a General Bikeshare Feed Specification (GBFS) API maintained by the city's technology provider (Lyft 2026). The feed includes data both in English and Spanish but in reality has inconsistent formatting. Because GBFS reflects only the current state of ECOBICI, the thesis uses trip-level data from the monthly CSVs and the GBFS API to reconstruct the map of stations by active status.

2.2 Data Preprocessing

In total, 96 files spanning three Mexico City administrations were preprocessed to ensure they had all necessary data and adhered to a standard structure. These files contained inconsistencies in naming conventions, language, and formatting. For this reason, a Python script has been prepared to parse data, harmonize column names, remove accents, and map Spanish-English variants into a single column map. Cleaning functions detected negative trip durations, trips exceeding 24 hours, or invalid timestamps. These files were then concatenated into yearly .parquet files for a combined total of 89 million rows (4 to 20 million rows each). The main data quality issue involved the month of Sept. 2022, which had 23% missing timestamps. All other data quality issues were resolved during this process.

2.3 Station-Hour Panel

Trip-level parquet files were aggregated into a station-hour panel using Raasveldt and Mühleisen (2019)'s DuckDB for large-scale data processing and memory efficiency. The resulting station-level panel includes: departures, arrivals, net flow, and absolute net flow. Additional temporal features include year, month, weekday, weekend, and binary variables for morning and evening peaks, which do not exist in the raw dataset. The panel structure is used for the exploratory data analysis and modeling process.

2.4 Exploratory Data Analysis

Descriptive statistics indicate that half of stations record a median of two departures and arrivals per hour. 75% of stations register five or fewer trips, while only a small number of stations reach maximums of 335 departures and 344 arrivals per hour. This means that the distribution is strongly right-skewed, since a small number of stations concentrate trip demand and pull the mean (3.61 of departures per hour) upward.

BSS demand. Figure 2.1 shows ECOBICI's growth trajectory was interrupted by the COVID-19 pandemic, followed by a sharp recovery and expansion phase both at the trips and station-level. Total annual trips fell from approximately 9 million to under 4.5 million during the COVID-19 years, recovering gradually until the first and second expansion phases, when the system surpassed 10 and 20 million trips.

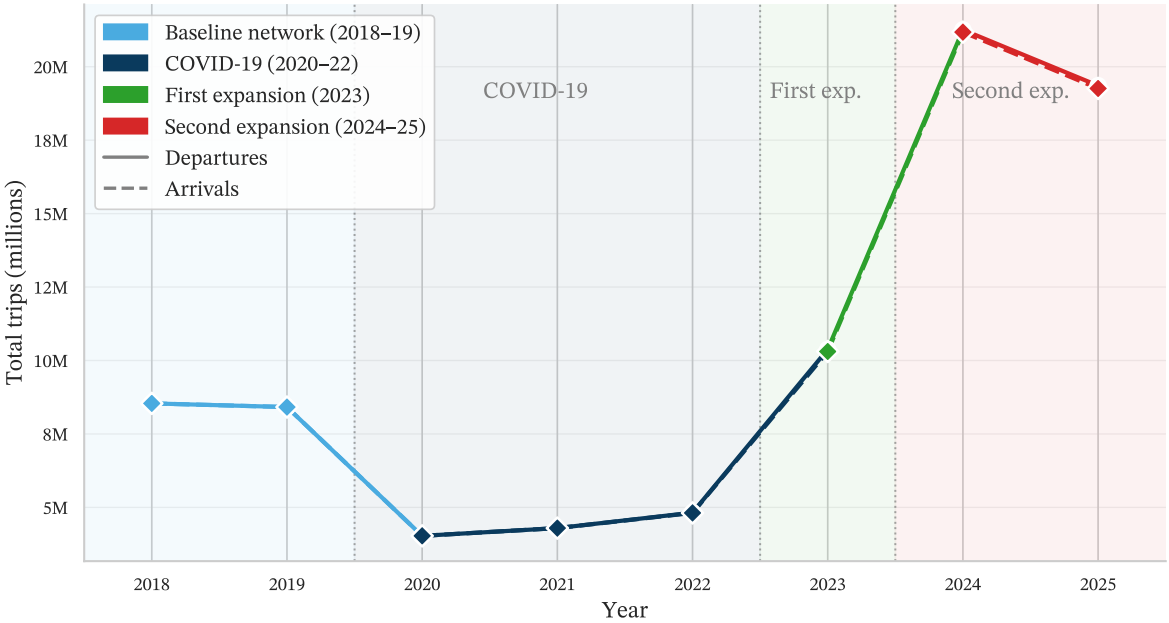

Figure 2.1: Total departures and arrivals by ECOBICI expansion

Station expansion. Figure 2.2 shows that demand growth takes place in parallel to network expansion, which grew from 482 baseline stations in 2018 to 666 active stations in 2025. Baseline stations (blue) concentrate in the center of the city. Between 2018 and 2022, the network remained static, and amid COVID-19, only two stations were added. Later expansions added 187 stations in 2023 (green) and 44 more in 2024 and 2025 (red), consolidating new service areas in the northern and eastern suburbs. Additionally, not all stations have remained active. Throughout this period, 53 stations were discontinued, with the highest turnover (34) taking place in 2025.

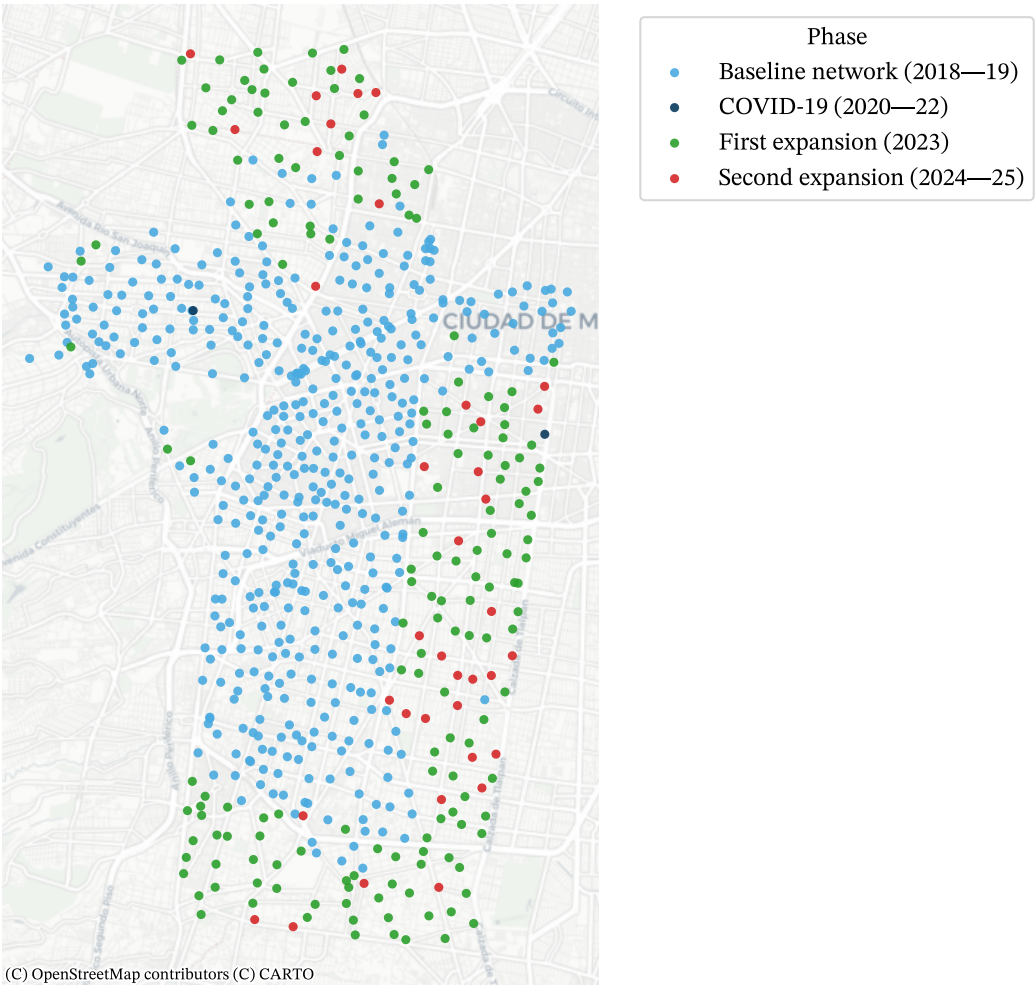

Figure 2.2: Spatial distribution of ECOBICI station expansion by phase

Weekday and hour patterns. Figure 2.3 shows that weekday departures concentrate around 08-9:00h and again between 17-19:00h (darkest blue), consistent with commuter flows in a typical work week. The intraday structure of demand has a bimodal pattern, as in other cases identified in the literature (Kaiser et al. 2024). Weekend demand has a flatter distribution, peaking slightly around midday (lighter green or turquoise), potentially suggesting a shift from commuter to leisure purposes.

Annex 5.3, Figure A.1 shows an hourly analysis of this bimodal pattern, which decreased during the COVID-19 years but remained stable across most other periods. These bimodal peaks also follow commuter flows (08-9:00h and 17-19:00h), with a slow recovery that accelerated from 2022 up to 2024. The only notable difference is 2025, in which departures remain high (around 1.4 million) but with more nuanced commuter peaks across the day, suggesting either changing user behavior or the impact of recent station closures (34 in 2025) on commute corridors.

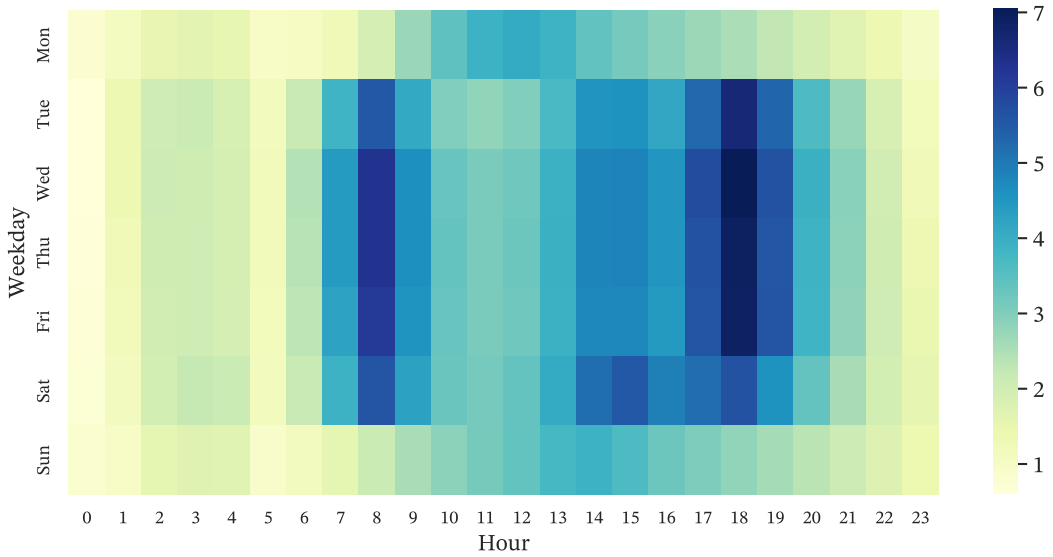

Figure 2.3: Distribution of average station departures by hour and weekday

Correlation of main features. Figure 2.4 highlights the moderate correlation between departures and arrivals ($r = 0.61$), suggesting that high-demand stations function as bilateral hubs rather than sources of bike supply only.

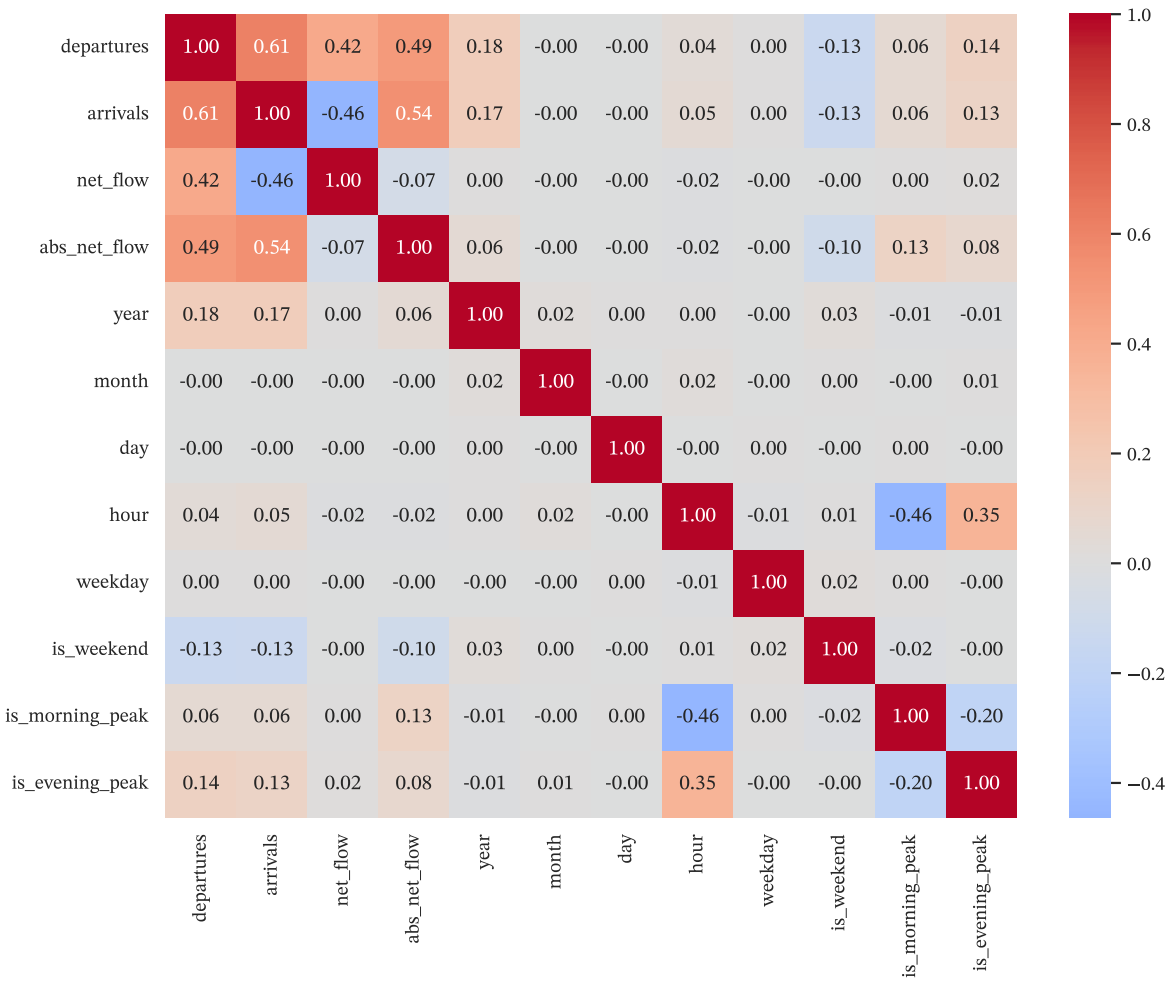

Figure 2.4: Correlation heatmap of station-level trips by hour

Arrivals and net flow are negatively correlated (r = -0.46), indicating stations with high incoming trips tend to accumulate surpluses. Morning peaks correlate negatively with hour (r = -0.46), reflecting the concentration in early commuting, while evening peaks do so positively (r = 0.35) with the end of work day. Calendar variables show little correlation with trip volume; special events and holidays are not explicitly included in this form.

Station-level net flow. Figure 2.5 maps the difference in net flow between arrivals and departures. Potential deficit stations (red) cluster around the city center, an area with corporate offices and government buildings, whereas surplus stations (purple) concentrate in the city's northern and southern suburbs. Weekends show fewer extreme deficits or surpluses, suggesting a more balanced and varied bike use.

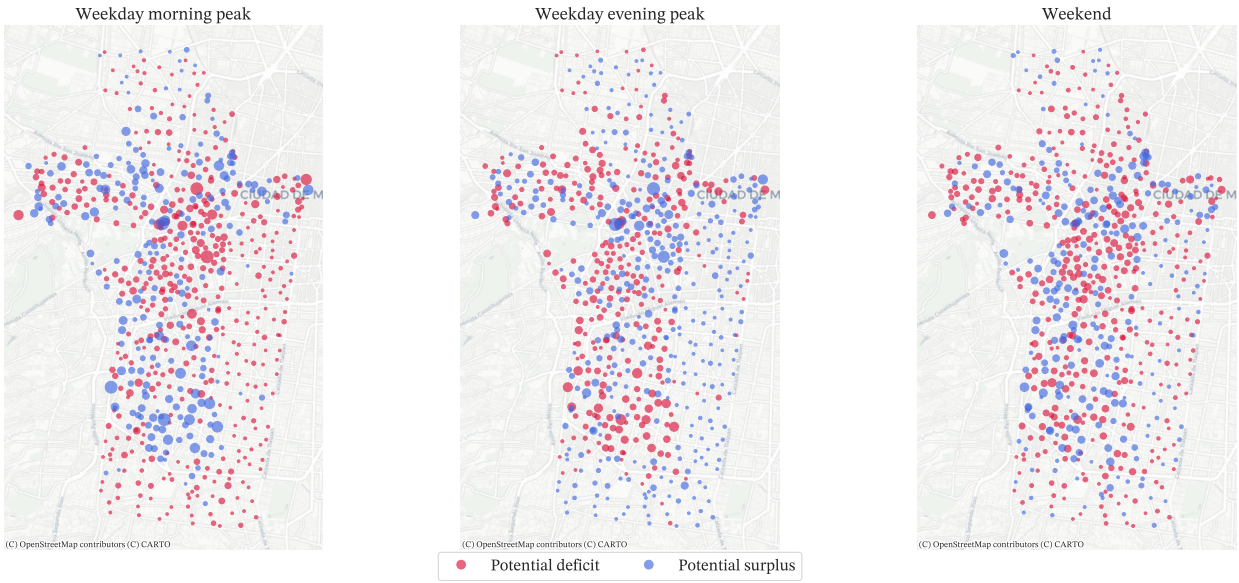

Figure 2.5: Station-level net flow by peak period and weekday, 2018—2025

Hyper concentration. Figure 2.6 identifies stations that disproportionately drive deficits and surpluses by applying a 95th percentile threshold (q = 0.95) to the net flow distribution. Based on this filter, the top 5% stations account for 20% of imbalance (sum of deficits and surpluses) but only 13-14% of demand during the morning and evening peaks. On weekends, this pattern suggests that demand is more evenly distributed, since the top 5% generate 17% of demand but only 12% of absolute imbalance.

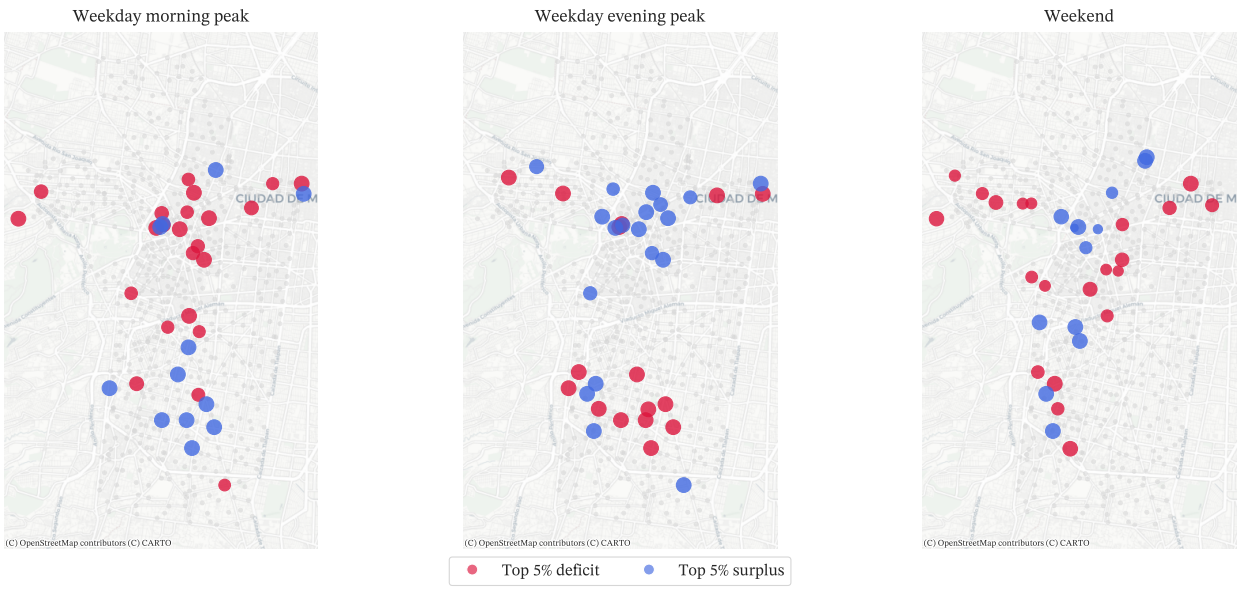

Figure 2.6: Quantile-based (5%) station deficit and surplus by time period, 2018—2025

To analyze whether high-imbalance stations have a structural persistence across years, a Jaccard similarity analysis compares the identity of top 5% stations before and after the COVID-19 pandemic. During expansion years (2023—2024), similarity of stations driving imbalance (J = 0.53-0.50) fluctuated, with new stations potentially disrupting trip arrivals and departures. However, 2025's recovery (J = 0.70) supports the insight that a subset of stations drive the imbalance pre- and post COVID-19, even despite station closures.

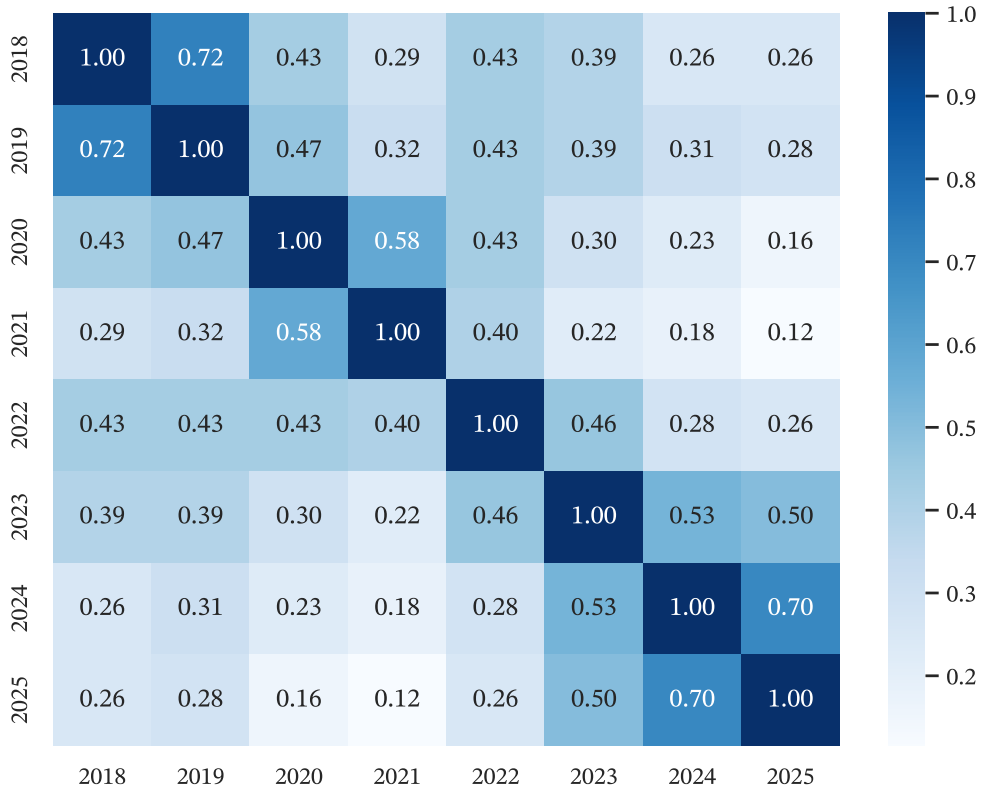

Figure 2.7: Jaccard similarity of station-level imbalance, 2018—2025

Chapter 3

Methods

3.1 Modeling Basis

This thesis addresses BSS rebalancing as a two-sided problem. First, model choice is treated as the first response to skewness for prediction and before any prioritization. From the exploratory data analysis, station-hour demand is highly right-skewed, with 75% of station-hour observations recording five or fewer trips per hour.

In the first part of the problem, machine learning models are trained to predict station-level demand, defined as departures per station per hour. The models include linear regression as a baseline benchmark, Poisson regression, and several tree-based methods. The assumption is that station-level demand and its predictors are unlikely to be purely linear. Tree-based models are therefore well suited to capture nonlinear relationships and handle outliers because they partition the feature space into non-overlapping regions and assign predictions based on the average response within each region (James et al. 2023), whether these be temporal, spatial, weather, and accessibility features.

Second, these predictions are used as inputs into a station prioritization framework designed to inform rebalancing across the BSS. Although this study does not directly focus on routing or mobility infrastructure planning, service level is widely used in the literature as a proxy for user satisfaction (Raviv et al. 2013; Maya-Duque et al. 2022). In this context, it is not enough to know whether imbalance concentrates spatially, but rather whether a subsample of stations drives these deficits and surpluses. Informing how to reduce the imbalance in these specific clusters could inform interventions aimed at improving bike availability across the BSS, and ultimately, make the system more reliable and efficient.

3.2 Feature Engineering

To prepare the dataset for demand prediction and prioritization tasks, additional temporal, spatial and contextual features have been added to the station-level panel:

- **Spatial data**: Station coordinates (latitude and longitude) were merged with trip records based on station id. Historical data shows that from a total of 719 stations, only 666 remain active in 2025 based on the most recent GBFS data. Only currently active stations were retained, excluding 53 discontinued stations.

- **Weather data**: Previous research has shown that incorporating rich weather variables improves accuracy and predictive performance (Ashqar et al. 2020; Kaiser et al. 2024). In this case, meteorological variables including temperature and rain were integrated based on historical records (Zippenfenig 2023).
- **Holiday data**: In Mexico City, congestion is heavily influenced by commuting patterns. Further, nearly half users commute to their workplace (Mexico City Secretary of Mobility 2024). Therefore, public holidays have been included (Murza et al. 2026), along with binary indicators for weekdays and weekends.
- **Multimodality data**: A spatial feature for multimodality has been constructed using the Euclidean distance from each ECOBICI station to the nearest metro, calculated by merging station coordinates from the GBFS feed with metro locations from the OSMnx package (Boeing 2025). This proxy for accessibility is justified, given that 56% of ECOBICI users combine their trips with other transport modes (Mexico City Secretary of Mobility 2024). The feature engineering process revealed that 50% of ECOBICI stations are located within 600–700 meters, and 80% are within 1 kilometer.

3.3 Prediction Model Framework

3.3.1 Target Variable

The target variable is departures at the station level per hour. In this case, departures are assumed to depend on station-specific, temporal patterns, and lagged demand, in addition to rich contextual variables such as weather, holidays, and multimodal accessibility. This relationship can be expressed in general form as:

$$d_{i,t} = \alpha_i + \pi_t + \chi_1 d_{i,t-1} + \beta_1 \text{Weather} + \beta_2 \text{Holidays} + \beta_3 \text{Multimodal} + \varepsilon_{i,t} \tag{3.1}$$

where:

- $d_{i,t}$: departures at station i at time t
- α_i: station fixed effect
- π_t: time effect
- $\chi_1 d_{i,t-1}$: lagged departures
- β_1: effect of weather conditions
- β_2: effect of holidays
- β_3: effect of multimodal accessibility
- $\varepsilon_{i,t}$: error term

It should be noted that this is a conceptual representation of demand for illustrative purposes, noting how deficits and surplus could depend on station and time effects and other covariates. Tree-based models do not impose this functional form since they learn these relationships flexibly.

3.3.2 Model Selection

Since departures are more stable than net flow but may still be unstable, multiple model types are implemented to capture patterns in the BSS. To this end, a comparative modeling framework is adopted, combining both statistical and tree-based machine learning methods. Linear Regression serves as the baseline benchmark, while Poisson Regression is a count-based alternative (Gómez Márquez et al. 2021). Tree-based models (Decision Tree, Random Forest, XGBoost) should handle the skewness and outliers in the distribution. The selection illustrates the broader trade-off between interpretability and flexibility in statistical learning from James et al. (2023) as illustrated below.

Figure 3.1: Model selection trade-off in statistical methods. From "An Introduction to Statistical Learning with Applications in Python" by James et al. 2023.

Prior research has shown that Random Forest or boosting models improve predictive performance when using heterogeneous data sources and feature-rich environments by reducing variance and learning from residual errors (Ashqar et al. 2020; Kaiser et al. 2024). The final set of models evaluated in this study is summarized in the following table:

Table 3.1: Model selection and purpose

Model	Selection purpose
Linear	Baseline benchmark approximation
Poisson	Count-based alternative for non-negative outcomes
Decision Tree	Nonlinear prediction via stratification of feature space
Random Forest	Ensemble model reducing variance through multiple trees
XGBoost	Gradient boosting estimator with regularization

Note: Adapted from Pedregosa et al. 2011, Gómez Márquez et al. 2021, James et al. 2023.

3.3.3 Training and Validation

Under a standard machine learning approach, this thesis would assume observations to be independent and identically distributed (i.i.d), with models using random splits. However, as shown in the exploratory data analysis, station-level demand depends on peak periods. Due to this, the thesis proposes time-based splits to prevent data leakage and assess temporal stability. The training set includes all observations from 2018 to 2024, while the test set includes observations from 2025 only.

3.3.4 Hyperparameter Tuning

Prediction Model. Three model classes were evaluated with different strategies.

- **Linear Regression:** No hyperparameter selection (baseline benchmark).
- **Poisson Regression:** Grid search with regularization (alpha: 1, 5, 10, 20) and convergence iterations (`max_iter: 300`). The `tune_poisson()` function evaluated all four configurations. The optimal set by RMSE was also identified (alpha=20).
- **Tree-based models:** Fixed hyperparameters due to computational constraints. Decision Tree (`max_depth=12, min_samples_leaf=50`) trained on full dataset. Random Forest and XGBoost (`n_estimators=200, max_depth=6`) trained on 10% sample of 2.1M observations (Chen and Guestrin 2016).

3.3.5 Evaluation Metrics

The approach evaluates model performance through Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which have previously been used in BSS-related research. MAE quantifies the average magnitude of prediction error, while RMSE penalizes larger deviations more strongly, considering sensitivity to outliers present in high-demand stations. Consistent with prior research using MAE and RMSE for bike prediction tasks (Ashqar et al. 2020; Miah et al. 2023; Kaiser et al. 2024), both metrics are used to capture variability due to the right-skewed distribution of station-hour demand.

3.4 Prioritization Framework

The prioritization component is implemented as a priority-scoring algorithm. Since about 5% of stations account for 20% of departures imbalance, while driving only 13-14% of demand, the target is this subset rather than the entire BSS. This acknowledges that system rebalancing is both computationally intensive and difficult to implement dynamically. For these reasons, the proposed approach does not aim to solve the routing problem.

Based on the research by Maya-Duque et al. (2022), which analyzes loans and returns of bikes as transactions in a BSS, station priority is calculated as:

$$p_i = \frac{T_i B_i}{\sum_{j \in E} T_j} \tag{3.2}$$

where:

- p_i: priority score of station i
- T_i: demand intensity at station i (total transactions: loans and returns)
- B_i: imbalance at station i (absolute difference between loans and returns)
- $\sum_{j \in E} T_j$: normalization across all stations in set E

In this thesis, the framework is adapted to use predicted demand and predicted absolute imbalance. Specifically, predicted departures capture expected demand pressure, while predicted absolute imbalance should show potential supply–demand mismatches and potential deficits within each time period. For the purpose of this implementation, predicted demand is obtained from the best-performing model (XGBoost).

3.4.1 Algorithm Adaptation

The following algorithm, adapted from Maya-Duque et al. (2022), should be interpreted as a prioritization heuristic rather than a full optimization model, such as those for routing. The score ranks stations by combining predicted demand intensity and predicted imbalance magnitude within each time period:

$$p_i = \frac{d_i B_i}{\sum_j d_j} \tag{3.3}$$

where:

- p_i: priority score for station i
- $d_i = \sum_t \widehat{d}_{i,t}$: total predicted departures for station i during the period

- $B_i = \sum_t |\widehat{d}_{i,t} - a_{i,t}|$: predicted absolute imbalance, defined as the sum of hourly differences between predicted departures (from the best model) and observed arrivals
- $\sum_j d_j$: total predicted departures across all stations in the selected period

The algorithm is applied separately to three temporal windows identified in the exploratory data analysis: weekday morning peaks, weekday evening peaks and weekends. For each period, the algorithm considers station-level metrics such as predicted deficit, predicted surplus, absolute imbalance and maximum hourly deficit. For each period, stations are ranked by the priority score and then classified with a 95th percentile threshold to identify those as high priority for potential rebalancing.

The number of bikes needed is estimated as a capacity-constrained peak deficit. The algorithm identifies the maximum hourly predicted deficit for each station, thus capturing peak deficit conditions rather than average behavior. This value is then capped by station capacity, so that the recommended number of bikes never exceeds the physical capacity of the station:

$$b_i^* = \min\left(\max_t \left[(\widehat{d}_{i,t} - a_{i,t})_+\right], C_i\right) \tag{3.4}$$

where:

- b_i^*: estimated bikes needed at station i
- $(\widehat{d}_{i,t} - a_{i,t})_+$: positive predicted deficit at station i and time t
- $\max_t(\cdot)$: maximum hourly deficit within the time period
- C_i: capacity of station i

3.4.2 Imbalance Evaluation

Observed imbalance is not included in the priority score itself, but is used afterward as an evaluation benchmark to assess whether prediction-informed prioritization, from machine learning methods, can better identify stations that are structurally imbalanced in the observed data. The evaluation is implemented by comparing the share of observed imbalance captured by the selected high-priority stations against baseline benchmarks.

Chapter 4

Results

4.1 Model Comparison

Table 4.1 compares model performance on the 2025 test set using the error metrics that evaluate which model has greater accuracy of predicted demand to actual values. As expected, and consistent with prior literature, tree-based models achieve the lowest MAE and RMSE, with XGBoost performing best on both metrics. This suggests that nonlinear models are better able to approximate station-level demand than linear or count-based alternatives. However, all models underestimate total departures, indicating a systematic downward bias rather than only random prediction error.

Table 4.1: Demand prediction model performance on the 2025 test set

Model	MAE	RMSE	Total (True)	Total (Predicted)	Bias (%)
XGBoost	**2.285**	**3.441**	18,658,166	17,118,718	-8.25
Decision Tree	2.308	3.475	18,658,166	17,223,653	**-7.69**
Random Forest	2.403	3.635	18,658,166	17,138,545	-8.14
Linear Regression	2.466	3.696	18,658,166	16,920,626	-9.31
Poisson Regression	2.964	4.716	18,658,166	14,393,569	-22.86

Even XGBoost, the best-performing model by MAE and RMSE as highlighted above underestimates total 2025 departures by 8.25%. This reflects the highly skewed distribution identified in the exploratory analysis. Most station-hour observations (75%) have fewer than five trips, while a small number of station-hours reach extreme peaks above the max. of 300 trips. An interesting finding is how the Decision Tree model has a lower aggregate bias than other tree-based methods, which could be due to how it predicts by stratifying the feature space based on local averages.

In this sense, it limits the usefulness of aggregate predictions for rebalancing, because the most relevant observations are precisely the peak-demand cases. Linear models were expected to underperform due to their lack of flexibility. A model can perform well on aggregate error metrics while still underestimating the demand peaks that are most relevant for rebalancing. XGBoost performance should be considered as conservative and not flexible enough to learn extreme or outlier patterns that concentrate departures.

4.2 XGBoost Performance

Figure 4.1 analyzes the relative prediction error by comparing XGBoost (best performing model) and Decision Tree (less biased model). In case of the prediction error share, the difference between actual and predicted departures has been scaled by actual demand. Positive values indicate underprediction and negative values indicate overprediction.

The similarity between both models suggests that this bias is not model-specific but demand-dependent: low-demand stations are overestimated, medium-demand stations are unbiased, and the highest demand stations (in this case those in the top 5% quantile) are consistently underestimated. XGBoost has a smoother and more concentrated bias pattern, whereas Decision Tree has greater variability at the tails.

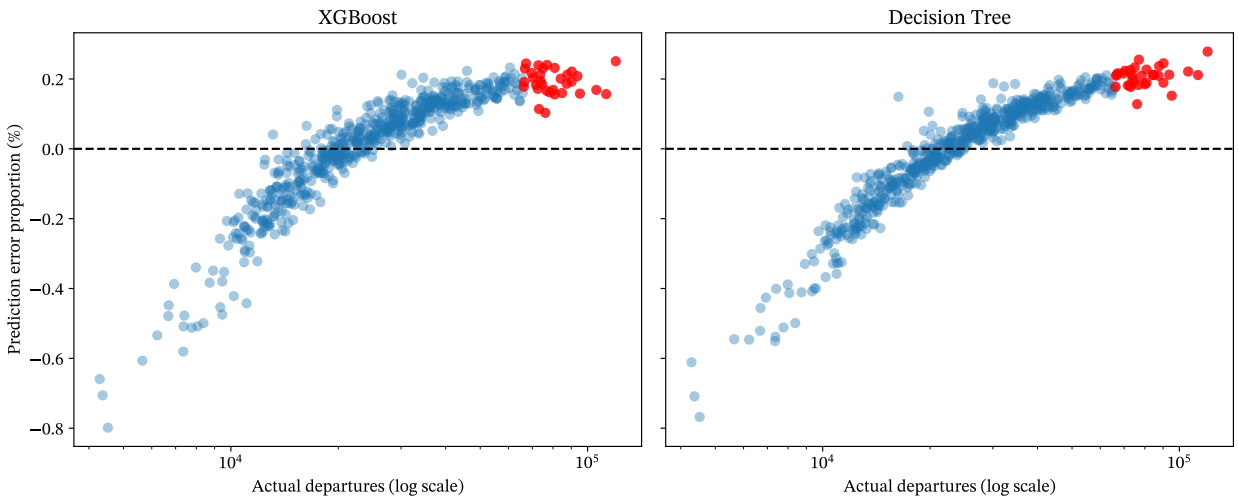

Figure 4.1: Comparison of station-level prediction error performance, 2025 test period

This pattern indicates that models tend to fit most of the demand distribution while failing to accurately capture extreme observations. As a result, even XGBoost is not capturing nonlinear demand patterns and provides conservative predictions for high-demand stations and underperforms in peak periods, as will be analyzed below.

4.3 Temporal Performance

Figure 4.2 confirms that temporal performance fluctuates according to the hour of day. Both XGBoost and Decision Tree show sharp increases in MAE and RMSE during the morning (07–9:00h) and evening (17–19:00h). By comparison, errors decrease sharply during off-peak periods. Consistent with BSS literature and the exploratory analysis, the pattern has a bimodal demand structure driven by commuting flows. Departures are driving these temporal peaks, and thus become more volatile and harder to predict.

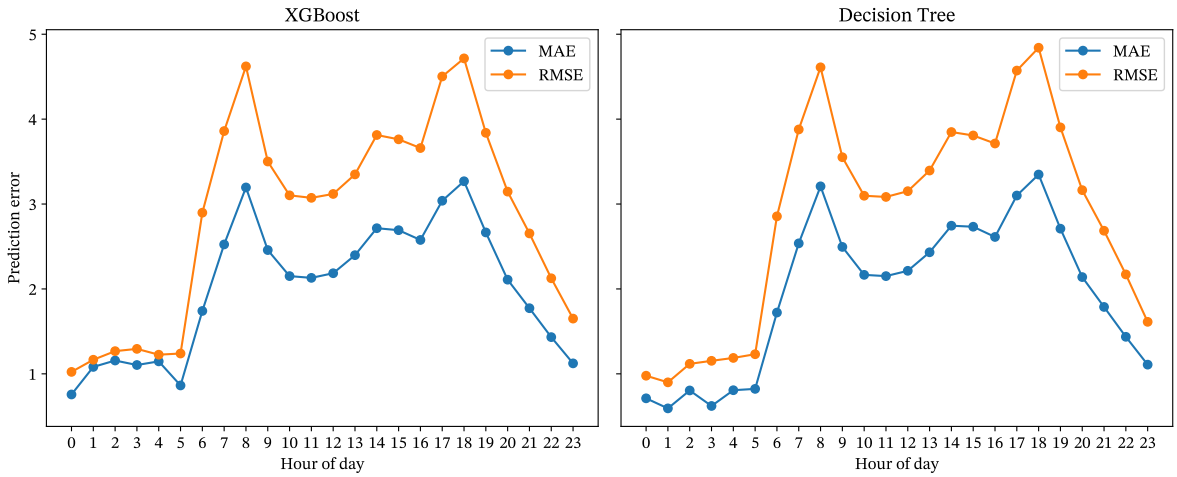

Figure 4.2: Comparison of hourly prediction errors, 2025 test period

MAE and RMSE behave differently across these periods. While both increase in a bimodal pattern, RMSE rises more sharply, particularly in the evening. Trained on historical data, a model should anticipate that these peaks will take place at a given hour, but it struggles to know the magnitude. This limitation is relevant for stations driving most of the imbalance, since models may capture the structural patterns of demand, but be less effective at predicting the actual departures of these peaks.

4.4 Feature Importance

Figure 4.3 compares features importance across the tree-based models, which should have identified nonlinear patterns but did not perform so well with high-demand stations.

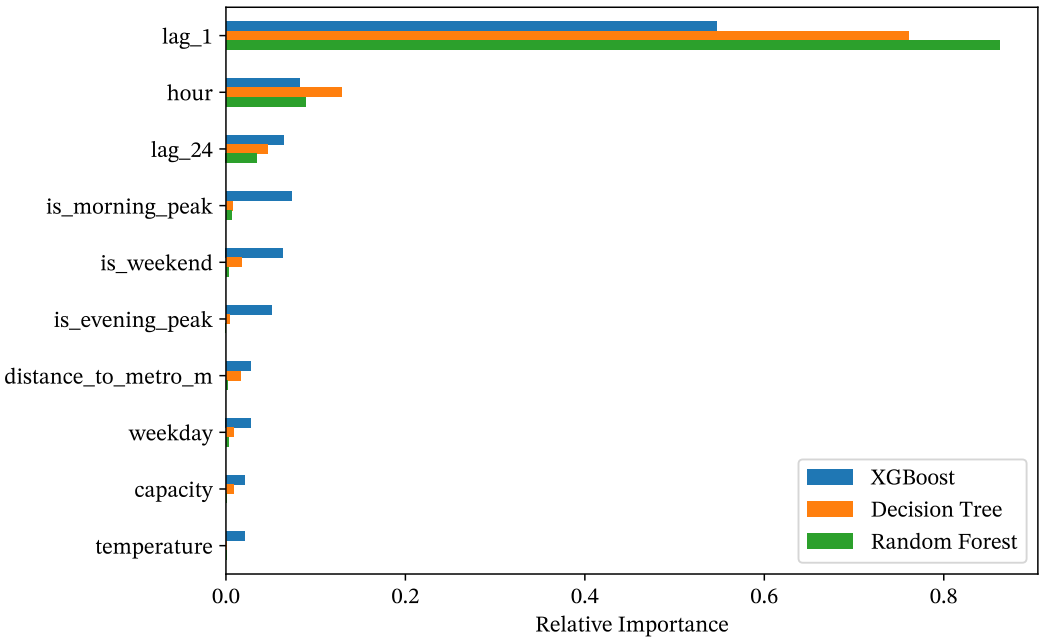

Figure 4.3: Feature importance for demand prediction across tree-based models

Lagged demand dominates as the feature with most relevance across all models. This indicates that it is the primary driver of station-level demand prediction. Temporal features are more important for the Decision Tree model regarding the hour of day, which can help explain why it is less biased than other models, given the concentration during peak periods. In contrast, other temporal or contextual variables, such as those by weather conditions, played a limited role. Surprisingly, the Random Forest model disregards key features entirely, which will be addressed in terms of its implementation in the discussion.

4.5 Prioritization Distribution

Figure 4.4 shows that priority scores are highly right-skewed, consistent with the exploratory data analysis. The majority of stations received low priority scores, as expected, while only a small subset forms a long right tail ranking as high priority. Scores above 40 are rare and correspond to the outlier stations driving system demand. The similarity across the peak periods also suggests that this has temporal patterns. Overall, the prioritization is behaving as intended by ranking a small subset, however, further analysis is needed into its relevance and comparison with observed imbalance in the 2025 test period.

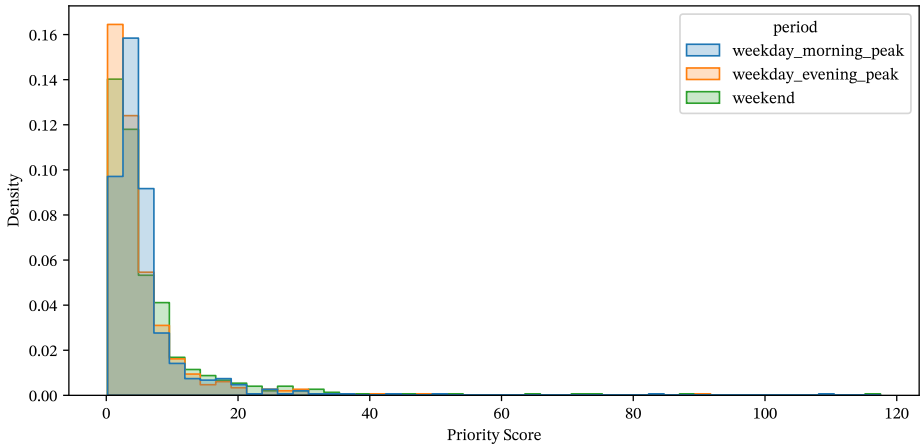

Figure 4.4: Distribution of station-level priority scores, 2025 test period

4.6 Weekday and Peak Period Relevance

Figure 4.5 compares priority scores with station-level net flow by peak period. Most stations cluster at low priority scores, while the middle region is empty and extreme net flow deficits and surpluses are concentrated towards the ends of the scale. As priority increases, scores above 20 are observed for a limited number of high-imbalance stations. In contrast, weekend periods show a tighter clustering with less extreme ends. Prioritization isolates this narrow group of high-impact stations, while most of the network remains concentrated in low to moderate imbalance regimes.

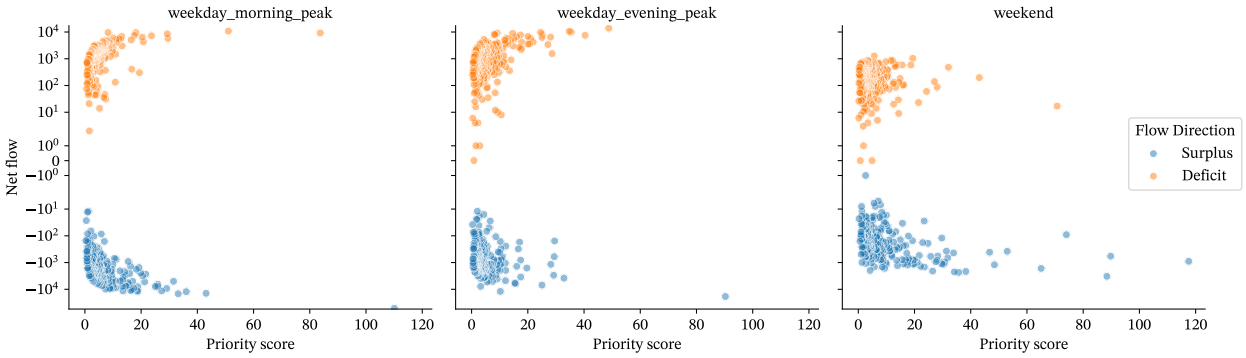

Figure 4.5: Net flow direction and priority scores by peak period, 2025 test period

4.7 Imbalance Mapping

Now, does prioritization, informed by predictions, improve or differ in terms of observed imbalance? A comparison shows that the top 5% of stations selected by the priority score capture 12.3% of predicted-informed imbalance, compared to 14.6% from the benchmark based on observed 2025 imbalance. However, the overlap between both stations is only 41.49%, indicating that the prioritization framework identifies a different subset of stations rather than reproducing the baseline. This could be due to underestimation or different feature selection from XGBoost.

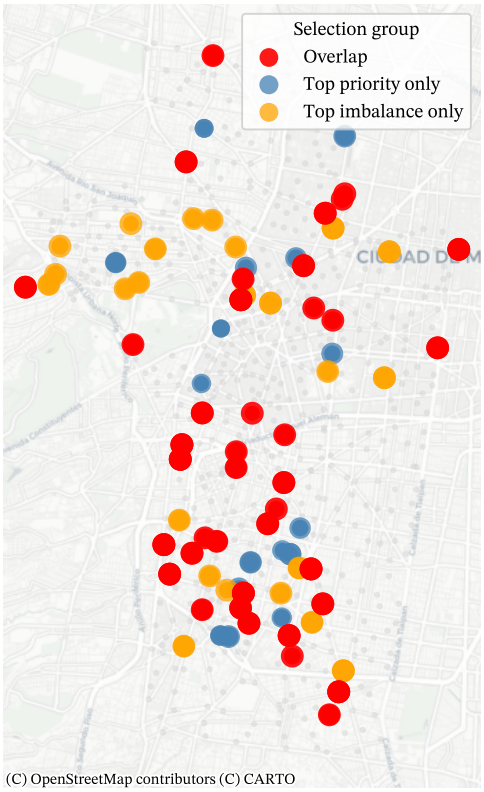

Figure 4.6: Spatial comparison of priority scores and observed imbalance, 2025 test period

Chapter 5

Discussion

5.1 Differences Between Models and Trade-Offs

The main proposition of this thesis is whether the past, seen through historical data, can be modeled to predict demand and inform priorities that will affect the future. The results presented aim to address this question, and while there are limitations in its implementation, the results suggest that predicted priority can make us rethink how cities use the data they already possess to guide rebalancing strategies. Cities are dynamic systems, and such social organisms require public goods accountability.

Prediction-based prioritization does not outperform a simple ranking of observed imbalance, yet it identifies a different subset of stations, introducing a different perspective for decision-making. Across prediction models, tree-based methods, particularly XGBoost, outperform linear and Poisson models in evaluation metrics such as MAE and RMSE. This is expected and confirms the presence of nonlinear relationships in station-level demand, but predictive accuracy does not translate to less bias. Models struggled with extreme observations and peak demand periods. Models performed well under stable conditions but errors increased precisely during those periods when the system faced stress.

The spatial and temporal analyses provide further insights into this difference. Stations tend to cluster by net flow: those with low priority on the ends of the tail, and as priority increases, outliers drive system imbalance. The implementation focuses on the best performing model, XGBoost, which underestimates the magnitude of imbalance. Its contribution lies less in improving predictive accuracy and more in challenging the assumption that historical imbalance patterns are sufficient under changing network conditions, where different policy stages and expansions have changed the system.

The prioritization, constructed from predicted demand and imbalance magnitude, seeks to anticipate the bike needs across the system. However, it too inherits prediction errors. As a result, the top 5% of stations by priority score capture 12.3% of total imbalance, compared to 14.6% captured by a baseline ranking based on observed imbalance. However, the overlap between both selections is only 41.4%, suggesting that prioritization is not replicating historical patterns, but instead identifying a different subset of stations.

5.2 Limitations

The implementation of the prediction models and prioritization framework reveals limitations in the design and approach that should be acknowledged in the context of this thesis.

First, the predictive models are heavily driven by time and peak periods, as reflected in the feature importance analysis. In this sense, modeling has identified gaps in the historical records of ECOBICI to integrate heterogeneous data sources, consistent with literature, that could improve predictions. Lagged demand dominates across all tree-based models, indicating that recent departures are the strongest predictor of near-term demand. While this is meaningful, the inclusion of heterogeneous features provided little predictive value, either because their effects are weaker or because they are absorbed by lagged demand. As a result, models perform well for typical observations but struggle to capture extreme demand, which is central to the imbalance problem addressed in this thesis.

Second, the largest limitation of modeling framework is that it remains static and does not capture the dynamic nature of BSS. While this limitation is common in the literature for diverse reasons related to real-world systems, other strands of literature propose simulation, game theory or deep-learning based models to simulate user behavior and system responses over time. This could have unearthed different patterns from the features engineered, such as weather conditions, station capacity or multimodality, which in the current state provided little predictive power.

In particular, hyperparameter tuning could have reinforced this implementation. The Random Forest and XGBoost models, for instance, were trained on a 10% sample of data to be able to run and constrained to shallow trees (with max. depth of six). By comparison, a simpler but more flexible Decision Tree model provided less biased results, likely because it identified localized patterns, such as hour of day effects, highlighting the trade-off between predictive accuracy and flexibility.

Finally, the prioritization component is intentionally simplified. The original intention was to predict demand and optimize bike rebalancing. This would have required detailed operational data, such as fleet size, staffing, routing constraints, or the use of simulated environments. Even existing studies using mixed-integer programming or simulation-based optimization are typically applied to small subsets of stations and acknowledge the computational challenges of scaling to real systems (Gómez Márquez et al. 2021; Maya-Duque et al. 2022). In this sense, the prioritization is a trade-off that aligns with observed imbalance, and does not presume to optimally solve the balancing problem. Instead, it provides a framework to jointly address the problem and inform strategies that may still be valuable under constraints.

5.3 Conclusions

Cities have a life of their own, with challenges rooted in historical patterns. BSS are no exception, and previous behavior can reveal why some of these patterns persist in time. This thesis has demonstrated that predictive accuracy does not guarantee superior prioritization: XGBoost models outperformed baseline methods in MAE and RMSE, yet identified a different subset of stations with only 41.4% overlap to observed imbalance. Models systematically underestimated demand during peak periods, exposing the limitations of static approaches. However, they point to an opportunity to apply deeper models—such as neural networks or game theory approaches—and different subsampling strategies to localize learning. The imbalances documented in this thesis are, in this sense, not merely operational puzzles but symptoms of access inequities that should be addressed.

Furthermore, the thesis points to the need for broader social dialogue between data infrastructure owners, system operators like ECOBICI, data analysts, and city officials involved in decision-making. Prioritization, optimization, and simulation-based solutions must address real-world constraints while making explicit the political choices embedded in allocation decisions. These choices must be interpretable to all involved: from the user taking or returning a bike, to the truck driver in charge of rebalancing, to the system administrator planning expansion. Limited resources can only stretch so far, making this a fertile field for technical innovation and, crucially, for genuine responsiveness to the people who commute and depend on these systems. Only then can cycling become not just efficient but algorithmically sound and socially just.

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Supporting Tables and Figures

Table A.1: Station trip share and imbalance metrics by period, 2018–2025

Period	Stations	Top 5%	Trip Share	Imbalance Share
Morning	718	36	0.139	0.208
Evening	718	36	0.130	0.176
Weekend	716	36	0.180	0.123

Note: Pre-modeling values include all stations with recorded trips.

Table A.2: Feature importance comparison across tree-based models, 2025

Feature	XGBoost	Decision Tree	Random Forest
lag_1	0.546	0.760	0.862
hour	0.082	0.129	0.089
lag_24	0.064	0.046	0.035
is_morning_peak	0.073	0.007	0.006
is_weekend	0.064	0.018	0.003
is_evening_peak	0.050	0.004	0.000
distance_to_metro_m	0.028	0.016	0.002
weekday	0.028	0.009	0.003
capacity	0.021	0.009	0.000
temperature	0.020	0.001	0.000

Note: Values are normalized and rounded for easier comparison.

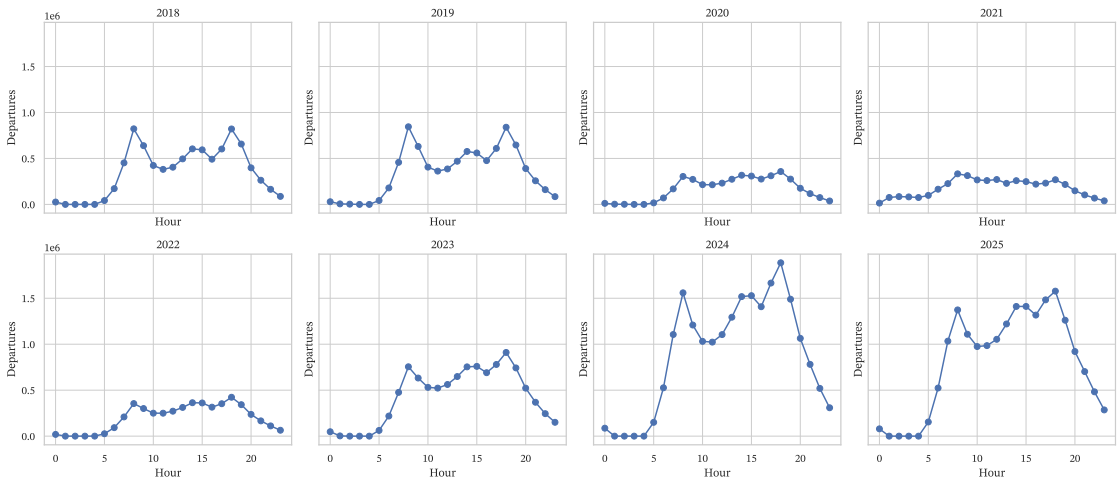

Figure A.1: Hourly station departures by year, 2018–2025

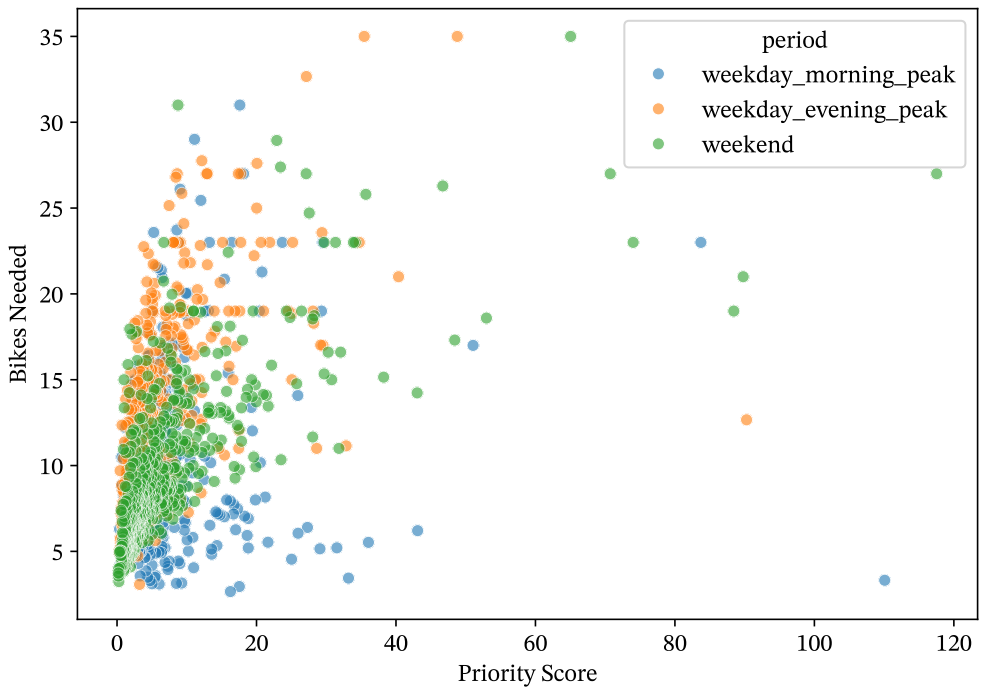

Figure A.2: Relationship between priority score and estimated bikes needed by peak period, 2025

Statement of Academic Integrity and Declaration of Authorship

I, **Franco Antonio Bastida**, hereby confirm and certify that this master thesis is my own work. All ideas, definitions, data sources and theoretical references authored by others are acknowledged in the text.

In accordance with the Hertie School's guidelines on academic integrity, I declare that I have used OpenAI's ChatGPT and Anthropic's Claude for code assistance, debugging and formatting and DeepL for language translation between English and Spanish and grammar support. All AI-generated content has been thoroughly reviewed, edited, and verified by me to ensure its accuracy.

I confirm that the digital copy of the Master of Data Science thesis that I submitted on April 30, 2026 is identical to the printed version submitted to the Examination Office.

Undersigns:

April 30, 2026
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